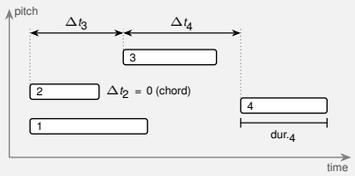
musicar 450M base (464M)

Pretrained multi-instrument MIDI model without text captions: 464.24M parameters (trunk 445.98M, cascade heads 16.28M, embeddings 1.99M), trained on 8.3B notes.

How to read: numbers give the reading order; teal = the model's structure; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom into the dotted part.

1 One note = one compound token

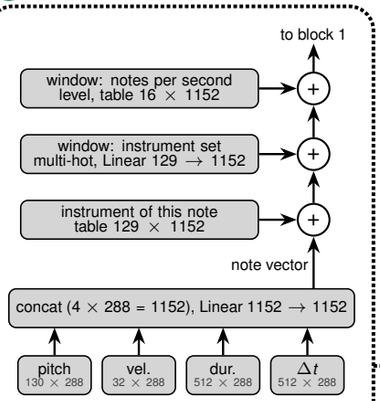


Notes 1–4 become positions 1–4: one position per note, in onset order, each with all five values:

field	values	meaning
Δt	512	time since the previous note started (20 ms steps)
instr.	129	General MIDI program 0–127 or drums
pitch	130	MIDI pitch 0–127 + start / end of a piece (BOS / EOS)
dur.	512	note length (20 ms steps)
vel.	32	velocity: how hard the note is played

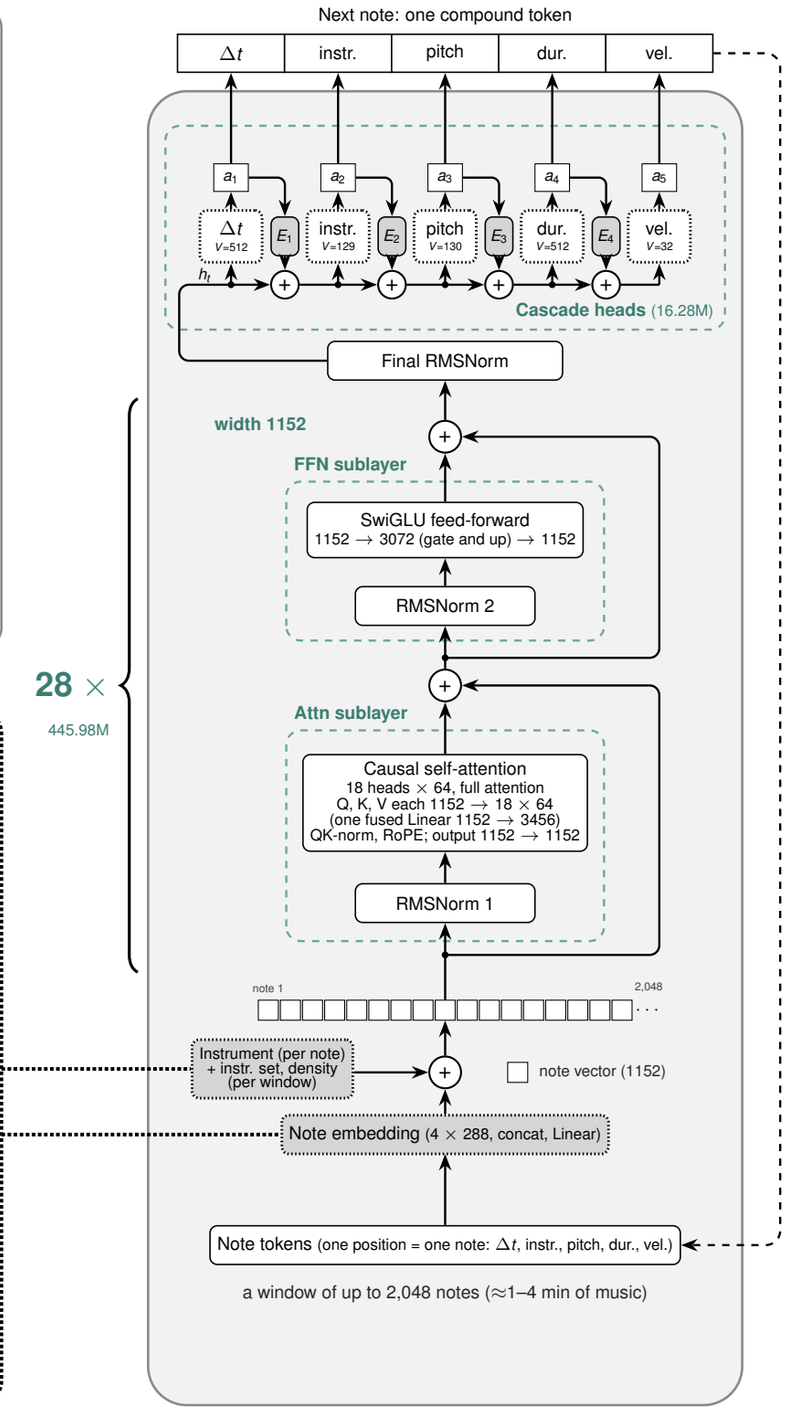
Time is absolute: no bars, beats or tempo tokens; Δt and durations stop at 10.22 s. The notes of all tracks are merged into one sequence; for drums the pitch is the drum sound.

2 Input embedding (1.99M)



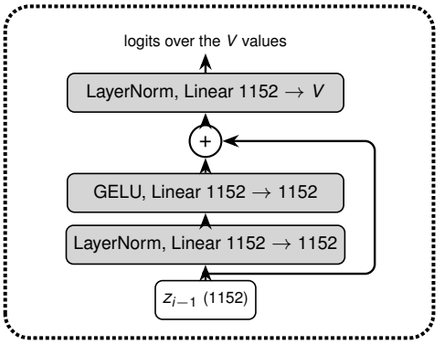
The instr. head predicts the next note's instrument (128 General MIDI programs + drums). The two **window conditions** are added at every position, measured on the window's own notes (density: 15 levels + "none"). In training both are replaced by "none" in 15% of the windows, so "none" (unconditional) stays usable; in generation they come from the user or the prompt. The instrument table and both conditions start at zero.

3 Music model



a window of up to 2,048 notes (≈1–4 min of music)

4 One cascade head (one per dotted box)



Why a cascade. One position predicts all five values of the next note. Head i reads the running state

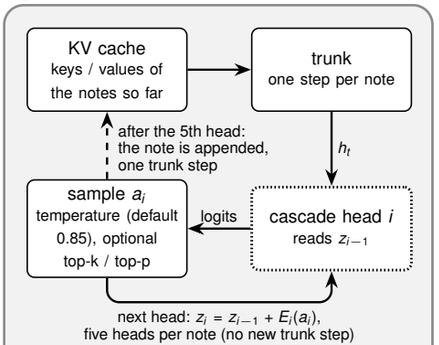
$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

the trunk state h_t plus the embeddings E of the values already chosen (tables 512, 129, 130, 512×1152; the last head needs none). So the note's probability follows the chain rule:

$$p(\Delta t)\, p(\text{instr.} \mid \Delta t)\, p(\text{pitch} \mid \Delta t, \text{instr.}) \cdots$$

Each head has its own weights. a_i is the true value in training (teacher forcing) and the sampled value in generation. Loss: the sum of the five cross-entropies. In a 42M test, the same heads without the chain were +0.34 nats / note worse.

5 Generation



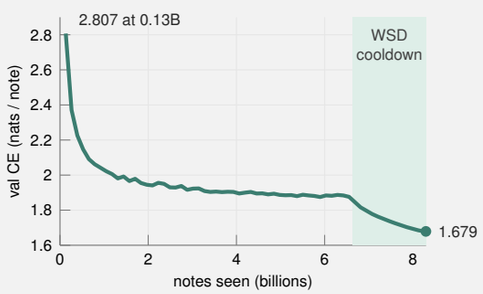
Start from BOS or from a MIDI prompt; EOS ends the piece. Past 2,048 notes the window slides by a quarter.

generate.py options

- `--prompt f.mid` continue a MIDI file
- `--sample-instruments` sample each note's instrument (else all piano)
- `--instruments 0,33,128` with it: the allowed programs (128 = drums)
- `--cond-instruments, --cond-density` the window conditions: a program list / notes per second, prompt, or none
- `--temperature, --top-p` all heads (T 0.85 by default)

6 Pretraining

Clean GigaMIDI validation (8.3B notes)



Every 2,000 steps (65,536 notes per step): the sum of the Δt, pitch, duration and velocity cross-entropies on GigaMIDI validation files whose notes do not occur in its training split. Constant LR, then the $1 - \sqrt{\cdot}$ cooldown in the last 20%. One run.

Setting	Base pretraining (done)
Data	GigaMIDI 0.4, Aria-MIDI 0.3, Discover MIDI 0.3 (sampling weights); no audio-derived songs
Notes seen	8.3B (≈1.8 epochs on average; Aria-MIDI ≈4.5)
Steps, batch	126,582 steps × 65,536 notes (32 windows of 2,048 notes)
Optimizer	Muon (Moonlight scaling) for the weight matrices of the blocks, AdamW for the rest; one LR 10^{-3}, weight decay 0.1
Schedule	WSD: 1,000 warmup steps, constant LR, then $1 - \sqrt{\cdot}$ decay to 0 over the last 20%
Loss	sum of the five cross-entropies, teacher forcing; no dropout, no label smoothing
Augmentation	random transposition by up to ±5 semitones (drums unchanged)
Conditions	instrument set and density replaced by "none" in 15% of the windows
Precision	bf16 with fp8 matmuls, torch.compile; the checkpoint holds bf16 weights
Hardware	one rented RTX 5090, ≈41 h
Result	clean GigaMIDI validation **1.679** nats / note = Δt 0.170 + pitch 0.367 + dur. 0.644 + vel. 0.497 (the instrument head, 0.117, apart); 1.681 with both conditions "none"